

g a research study by using a standardized questionnaire that was developed as a part of the Kano analysis model.

Designing a survey for a Kano study is rather simple from the research perspective, however it requires a lot of preparation by the Product Manager - especially describing the features and current state of the product area. In the following sections, we will discuss details for designing a Kano study at GitLab.

Kano Survey Example

In the project below, a Kano survey was developed to understand how to prioritize a set of new GitLab features.

Example project: Survey for CI feature prioritization

Project summary: The team was considering 12 new features to build that they thought would benefit our users. Since building new features is costly and time consuming, the team wanted to understand how users felt about each of those features, along with a prioritization. The goal was to end up with a clearer understanding of which 12 the team should build first.

In order to get feedback about the 12 features, the team needed to develop and iterate on a short summary or description for each of the features. The following sections provide details the process to create feature descriptions and transfer them to a survey.

Preparation

Well-formulated feature descriptions are crucial for the success of your study. Participants must be able to accurately understand the features that you are asking about and the value of those features, so they can provide honest feedback. It makes sense to begin creating the survey only after the features have gone through an in-depth refinement process to ensure they provide enough detail to be clearly understood.

Category description

We usually develop new features in the context of features that already exist. For that reason, you should first walk the participant through the current experience(s) that contain the features being presented.

Example of category description:

> A continuous integration pipeline in GitLab is defined as a version-controlled file in the Git repository of a project. This file is called 'gitlab-ci.yml' and users configure CI pipelines by editing this file and pushing it to the repository. Pipelines consists of jobs and stages. Jobs define what the pipeline should do. Stages consist of one or multiple jobs and define the sequence in which jobs run. All jobs in a stage run in parallel, and if all of them succeed, the pipeline moves on to the next stage. Different types of conditions can be specified for pipelines like prerequisites, exceptions, and so on. You can edit the pipeline YAML file either by cloning the project locally or using an online text editor.

Feature descriptions

When creating feature descriptions, follow these guidelines:

1. Describe features, not MVCs

- The findings should be relevant for a reasonably long time. In other words, you would likely not be able to develop it in 1 or 2 milestones.

1. Use simple and neutral language to avoid bias in how feature descriptions are perceived

- No GitLab lingo like sections, categories
- No sales or marketing words like easy-to-use, better, faster, more efficient
- Bad example: Run Dynamic Application Security Testing (DAST) in the cloud as part of your CI jobs on cloud based runners. This will make deploying and testing applications in a sandbox significantly more efficient.
- Good example: Run Dynamic Application Security Testing (DAST) in the cloud as part of your CI jobs on cloud based runners, to deploy and test applications in a sandbox for potential vulnerabilities.

1. Set the feature into the context of the system

- Use consistent terminology, making sure your feature descriptions fit into the overall category description.

1. Aim for one paragraph of text for each of the descriptions

- Less than 1 normal paragraph might be not enough. On the other hand, if the text is too long, participants will more likely skim through the content and not develop a full grasp of the feature.

1. Include only new features

- Including features that are already developed can set a benchmark; however, it is mostly not actionable and therefore useless in this context.

1. Enrich the descriptions with images or GIFs

- Having a visual aid can help participants understand the concept being described in the feature description.

In general, feature descriptions should follow this basic structure:

- Feature: What can the user see in the UI
- Behavior: What does it do
- Value: Why do we want to introduce it

Example of feature description:

> A list of CI job code snippets that will display next to the online pipeline editor. You use these code snippets as the building blocks of your pipeline by copying the jobs YAML from this list and pasting it into your pipeline code. Thus, there are fewer steps in the process of creating new pipelines.

- For more feature description examples, see this CI features list Google Doc.

Tips & tricks

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{{% alert title="· Tip" color="primary" %}}
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: Preparing for the survey can take a lot of time and might be hard to track. Creating a separate issue to track the effort can help break up the main research issue into smaller manageable parts.

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{{% /alert %}}
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{{% alert title="· Tip" color="primary" %}}

Creating feature descriptions asynchronously can take a lot of time. If you feel like your team is not moving forward quickly enough, pivot to synchronous communication.

{{% /alert %}}

Survey

Standardized questionnaire

The standardized questionnaire for the Kano model consists of two questions that are asked for each of the features that you are interested in:

Functional question	Dysfunctional question
How would you feel if you had this feature? - I am delighted by it - I expect it - I am neutral - I can tolerate it - I am frustrated by it	How would you feel if you did not have this feature? - I am delighted by it - I expect it - I am neutral - I can tolerate it - I am frustrated by it

The benefit of using the standardized questionnaire is that basically all of the research design is already done, and you can reuse it from previous studies. Also, taking a consistent approach to this methodology reduces the risk of introducing errors to the process.

For an example of a research plan, see this past Kano study plan in Google Docs

Questionnaire structure

Use Qualtrics to create the questionnaire. Each feature description with related questions

(functional and dysfunctional question) should be presented on a separate block, so participants can answer each feature independent of another. Also, it is recommended to randomize the order of each block, so the presentation order does not affect participants' feedback.

Example survey that we used in 1027

Data collection & Analysis

Using the provided questionnaire, you'll be able to collect the data, analyze it, and report out the findings using both quantitative and qualitative approaches.

The qualitative approach

This approach will provide you with a better understanding of the why behind their prioritization. Collecting and analyzing the qualitative data first can help you to spot problems in your feature descriptions, such as whether they are confusing or unclear, so you can fix the descriptions before sending them to a larger set of participants. After taking the quantitative approach, you can also talk to a few users from the survey sample for some additional feedback on why they provided their ratings.

We recommend conducting 5 to 10 moderated sessions and/or 20 to 30 unmoderated sessions where a participant's task is to go through the questionnaire and explain the reasoning behind their answers.

Example Dovetail project where moderated interviews were conducted before and after collecting responses via a larger survey.

The quantitative approach

We recommend collecting responses from 50 to 80 users in your target audience and analyze them using discrete analysis described in The Complete Guide to Kano Model. This approach will provide you with "the numbers" that can back up your prioritization decisions, but you will still lack the understanding behind the prioritization.

For the analysis use this spreadsheet template (GitLab internal link).

Example report (GitLab internal link) that summarizes the output from the quantitative approach.

Taking the quantitative and qualitative approach will result in a more complete story behind the data. In addition to prioritization, you'll also be able to explain why participants scored the way they did, too. Combine these two approaches to get the most of your Kano model study.

Additional resources

- Article: The Complete Guide to Kano Model
- Article: Kano Model · Ways to use it and NOT use it
- Video: Building a Winning UX Strategy Using the Kano Model

Examples of Kano research conducted at GitLab

- Monitoring feature prioritization
- CI feature prioritization
- Threat insights feature prioritization
- GitLab Plus feature prioritization

SECTION: product/ux/ux-research/longitudinal-studies/index.md

A longitudinal study looks at how something changes for the same group of people, over a period of time. Data is collected at predefined time intervals to determine how the measures change over the designated period of time.

When to use?

Longitudinal studies are perfect to help teams answer research questions that are associated with an element of time. Sometimes teams want to understand how perceptions, knowledge gain, etc, may change over time.

Some examples:

- How will users react to a new navigation scheme, even after using it over a period of time?
- How does confidence and product knowledge change for new users as they're starting to use GitLab during their first 60 days?

Which methods are used?

Longitudinal studies typically combine use of multiple methods. Determining which methods to use for longitudinal studies is based on the research question(s) that need to be answered. Essentially, you will want to choose methods that will best answer your research questions. Here are some examples of methods that can be used for a longitudinal study:

- Interviews - It's a best practice to start every longitudinal study with interviews that help to establish a relationship with your participants. This is particularly important in longitudinal studies, since you will have multiple touchpoints with each participant throughout the study. Interviews also allow you to set expectations about the study's cadence and timeline, while allowing the participant to ask any clarifying questions about their role.
- Diary studies - These are often incorporated into longitudinal studies, where participants are asked to document something they've done over a period of time.
- Surveys - It can be helpful to run surveys throughout the course of your longitudinal study, mainly as a way to revisit the key measures and determine how those have changed over time. Note that these can be done through a survey tool, like Qualtrics, or verbally through interviews.

To reiterate the importance of interviews: they're critical to understanding the 'why' behind the data that may appear in diary studies and the ratings that participants provide within surveys. The 'why' can directly lead to clear actions, too.

Touchpoints, or formal contacts with participants, are a necessary part of a longitudinal study. They're used for a number of reasons:

- To check in on participants
- As an opportunity to assess the key measures
- To dive in deeper on the ratings and data that the participant has provided to date

Measures

The measures used for longitudinal studies can vary greatly and depend entirely on the research question(s) being answered. However, always consider the factor of time when planning key measures, so you can demonstrate fluctuations to things like attitude or confidence throughout the course of your study. Examples:

- Confidence in using the product (1-7 Likert scale rating) - collected at Day 1 (control), Day 14, Day 30, Day 60
- Effectiveness in using the new navigation (1-7 Likert scale rating) - collected at Day 0 (control), Day 1, Day 14, and Day 30

The above examples use self-reported measures, but it's certainly possible to leverage usage data, instead, by tracking measures like error counts over time.

A sample output could look something like the following chart, which demonstrates how the measure changes over the designated period of time for a single participant:

Steps to conduct a longitudinal study

1. Ensure your research question(s) can best be answered through a longitudinal study.
1. Design an achievable longitudinal study by considering variables like: GitLab versions, permissions, personas, JTBD, GitLab and technical experience, etc. Among the variables you consider, determine what are the key measures too.
1. Think through a data management plan, as longitudinal studies yield a significant amount of data. This includes how you will store, organize, and eventually analyze the data.
1. Determine the timeframe during which the study will take place, along with the necessary amount and cadence of planned touchpoints.
1. Open a recruitment issue that outlines the number of touchpoints and time commitment for each one.
1. Determine the gratuities for each touchpoint.
1. During initial 1:1 interviews, work with participants directly to schedule the upcoming touchpoints.
1. Work with the UX Research Operations Coordinator to make sure gratuities are delivered in an efficient and timely fashion.

Challenges

Longitudinal studies have their own set of challenges that you should be aware of:

- They naturally take time to conduct — often months due to the time component involved.

- Since there are multiple touchpoints per participant, that translates to additional effort for the DRI conducting the research.
- Participant attrition can be high with longitudinal studies, so you may need to provide a higher incentive and/or recruit additional participants up front to account for participants you will lose along the way.
- Longitudinal studies can result in a significant amount of data. DRIs can become overwhelmed, if a data management plan isn't in place prior to starting.

Tips and tricks

Gratuities - A gratuity for each touchpoint is useful in keeping participants active. However, offering a reduced gratuity amount for the touchpoints preceding the final touchpoint is a strategy one can take - as long as the final gratuity is much larger. This makes up for the previous smaller gratuities and also acts as an incentive for participants to see the study all the way through. The table below shows an example of how to design a gratuity structure that has more of an incentive at the end of the study:

Touchpoint	Time scheduled	Gratuity amount
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1	45 minutes	\$30 USD
2	15 minutes	\$20 USD
3	15 minutes	\$20 USD
4	60 minutes	\$200 USD

Set clear expectations - Make it crystal clear to participants how important it is for them to complete the full study by taking part in all touchpoints. Otherwise, they don't qualify for future gratuities and they're removed from the study. Explain the purpose of the study, while stressing how important their participation is to the success of the research. You'd be surprised how invested they'll be in the project, if you share some details with them. Providing users the touchpoint schedule with dates can be helpful too.

Data management plan - Focus only on what's important. Typically, that means what is needed to answer the research question(s). It's okay to include some other data that supports the main insights, but set limits for yourself. It's also helpful to input/analyze data — either in real time (during the interview) or immediately after it. Creating templates can help with this. The key is to be highly organized and disciplined in what you choose to capture, when you capture it, and how you capture it.

Use Google Docs - If you're using a diary study approach, use Google Docs to create a template for the study. Then, use the template to create one document for each participant. Share the document with the participant and give them edit rights, so you have full visibility into their entries. If you feel they need a nudge, or if you have a question about one of their entries, you can just @-mention them in their document. At the conclusion of the study, be sure to unshare all documents, so that participants can't make additional changes.

Get the 'why' - Make sure to build in opportunities to understand the 'why' behind the data you're gathering. This is critical in telling a complete and impactful story.

Give yourself time - Remember that you'll need a lot more time to conduct a longitudinal study than a usability study. You'll also have a lot more data to look through and think about. Make

sure to allocate additional time for yourself, and clearly set expectations with your stakeholders.

Finding quality participants - Since a longitudinal study requires a longer time commitment, getting quality participants and feedback is essential. A strategy you can use when screening participants is to do an intro interview first before inviting them into the study. Use this interview to collect some meaningful information, while also assessing the quality of feedback the participant provides.

SECTION: product/ux/ux-research/mental-modeling/index.md

A mental model is what a user believes to be true about how a system works. This definition has two implications worth pointing out: a user's mental model may in fact be misaligned with the Designer's intent (or how the product is currently laid out), and that every individual may have a slightly different mental model given the same interface or product.

Nielsen/Norman Group states:

"Mental models are one of the most important concepts in human-computer interaction (HCI)", and "What users believe they know about a UI strongly impacts how they use it."

A mental modeling study aims to gather information on users' beliefs about how your product operates. They are especially common when dealing with information architecture and navigation.

When to do a mental modeling study

A mental modeling study can be done at any point in the product lifecycle, but it tends to be most impactful earlier on, when concepts are still in a more flexible state. A frequent use case is when designing or redesigning a product's navigation, such as menus and sub-menus, etc.

In actuality, mental modeling can be a useful exercise in clarifying any part of a product - or even the product as a whole.

Types of mental modeling activities

There are a couple of different exercises that can be used to do mental modeling, and many variations from those main types. We will go over the two main types on this page: card sorting and cognitive mapping.

Card sorting

Card sorting is a very common method in UX research for discovering how users might group a set of items, for instance in designing an information architecture. Card sorting reveals mental models by having the participant sort a deck of cards, each with a relevant term, into piles. This can be done in person with index cards, or virtually with a digital whiteboard tool (e.g., Figjam).

You will end up with a set of piles containing the cards or terms a participant believes go

together in some way. It is often helpful to have the participant give a name to each group, or to combine several groups under a new term they come up with - this is especially useful in determining multi-layered navigation models.

Depending on the flexibility of the space you are trying to model, you can leave some cards blank and allow the user to add their own terms into the mix. This can be a great way of discovering gaps in your own mental model, but it can also introduce noise into the results.

Cognitive mapping

Cognitive mapping, according to the Nielsen/Norman Group, is really an umbrella term for all visual representations of mental models. This link from them is a great resource to find out about the different flavors of maps (e.g., cognitive maps, mind maps, concept maps) and what each method is particularly useful for.

At its core, cognitive mapping involves having a user create a visual representation of the system or area of interest you're investigating. There are a few variables to consider when designing the mapping activity:

Structure

The context of your domain can help determine what kind of structure you use in your mapping exercise. For example, if you are focused on a strictly hierarchical space, such as exploring everything that might be under a menu item named 'settings', you may want to provide a visual representation that indicates levels of the menu structure. If it is something more abstract, perhaps a blank canvas is all you need.

Relationships

In card sorting, we only deal with 'nouns' - the objects or items we operate on. In mapping, we can also introduce 'verbs' - or the relationships between nouns. Verbs are typically represented by arrows (with labels) or are differentiated by shape from nouns (circles versus squares, for example). Gathering information about the relationships between objects can be extremely informative for studies aimed at understanding a more complex system, such as data flow through a system or organizational structure.

Vocabulary

You have the option of providing all the vocabulary to be used in the map, some of it, or none at all. Choosing the right set of words to provide will have a large impact on the outcome of your study. For our settings example, you may choose to have only that word. For a study on how people organize their refrigerators, some basic items or types might be provided. Other cases may demand a larger list of nouns and verbs, like understanding the relationships between different products as data travels through them.

It's important to reconcile these variables with the research goals you're after. A well-designed mapping activity can yield a great deal of insight into the mental models of your user, and can have a large positive impact on product planning and implementation.

Below is the set up we used for a recent mental modeling study using cognitive mapping. We

wanted to gather the ways in which customers organized the different kinds of work items they dealt with every day. Because we had specific nouns and verbs in mind, we pre-populated areas of each, but allowed participants to add their own as they needed.

Mental model analysis

Due to the variable nature of mental model studies (as discussed above), there isn't one formula for analyzing the results of a mental modeling activity. However, there are some common techniques and things to look for when breaking down the results of your study.

Card sorting analysis

Analysis for card sorting is fairly straightforward. You can look at the terms that appeared in the same piles as one another, and count them across participants. This will give you some numbers that indicate how 'close' the items are (called a similarity score). For instance, if term A and term B were always sorted into the same pile, they would have a 100% similarity score.

Additionally, if you collected names or terms for the groups of cards, you can compare those parent terms to each other (which group names surfaced more often, for instance), and compare which items fell underneath each group.

Concept mapping analysis

Analyzing concept maps is less cut-and-dry. The techniques available to you will vary based on the structure of your map. Here we will go over two methods: segmentation and network graphing.

Segmentation

There are a few ways to do segmentation, but essentially segmentation is about looking at the shape of the maps and finding clusters of items or nouns that tend to be grouped together, and that form a weak link to the rest of the map (defined as having few connections coming into or going out from that group).

When you see clusters like this that share common nouns and appear across many of the maps you review, it is an indication that a section of your map can be segmented in some way. It might be a particular workflow, a sub-process, or something else, but it is usually worth noting. For example, when examining concept maps of a product life cycle, you may find segments like 'planning', 'execution', or 'validation' that identify a part of the process.

Below is an example of the segmentation process, broken down into 3 steps. From left to right, we start with the raw map, clean it up, and then are able to see areas of the map that are isolated from the others (outlined in squares).

Network Graphing

For concept maps where both nouns and verbs (objects and relationships) are recorded, you can create a network graph. This is done by treating each noun as a node and each verb as an edge, and recording all the noun-verb-noun relationships. This involves going through your maps and tabulating all the unique relationships expressed.

What you end up with is something like this:

This is from the same mental model study done at GitLab on different Work Items. Each blue circle represents a node or noun, and the size of the circle corresponds to how many relationships it has with other nodes in the graph.

Once you have constructed your graph, there are some well-defined measures for interpreting the nodes within that graph:

- Connections: the number of connections a node has to other nodes.
- Relevance: the relevance of each node is based on its number of connections and relevance of the nodes it's connected to (whose relevance is, in turn, based on their connections).
- Betweenness: how many times a node is located along the shortest path between two nodes.
- Closeness: how close a node is to the other nodes in the network, or how many steps it takes from a node to reach the other nodes on average.
- Coreness: indicates the layer of the network a node belongs to. The layer with the highest index is the core of the network.

One can usually find a tool that will perform these calculations for you automatically, but the formulas are readily available through a simple internet search (onodo.org is a fairly accessible one.) Besides providing a visual understanding of your results, one of the benefits of performing these calculations is that you will come away with a deeper understanding of the prominent (and less prominent) objects across all the concept maps you collect, giving you the 'core' concepts or nouns related to your study.

SECTION: product/ux/ux-research/preference-testing/index.md

Preference testing help you choose between design variations (icons, one page mockups, color palettes, design components, and more) by asking users to select their preferred option. You can have the participant select a preference based on the appearance of the design, how well it communicates information, or other qualities that evaluate your hypothesis.

It is important to note that you should not use this method to test workflows, usability, or anything that requires interaction. It can be used in conjunction with an open-ended question to ask about why participants made their selections to understand the reasons behind preferences. A moderated format would work best for that; as the moderator can ask more follow-up questions.

Tips for creating Preference Tests

- Any visuals used must contain the full context for the design to be accurately assessed.

- Screenshots must be large enough to be clearly viewed and read easily.
- These tests are commonly used to measure aesthetic appeal, but participants can be instructed to judge designs based on their trustworthiness or how well they communicate a specific message or idea.

How to run a preference test in Qualtrics

Step 1: After logging into Qualtrics, click on "Create a new project" that appears in the bottom left of the side panel that is called "Welcome to XM".

Step 2: Choose to create a blank survey project

Step 3: Add your task scenario using the Text / Graphic question type

You can skip Steps 3 through 6 below by importing the "Preference Question Block" from the Questions folder in the UX Research & Product Question Library.

Step 3: Add your preference questions using the 'Multiple Choice' question type

- Make sure the 'Allow one answer' option is selected
- Click on the response item and select the 'Insert Graphic' item from the dropdown
- Upload the graphic from your computer by selecting the 'Upload a new Graphic' from your computer
- Repeat for each response option

Step 4: Add a question that will help you understand user preferences

- Follow that question with a Text Entry question to understand why participants chose that option.
 - Example questions:
 - Why did you prefer that design?
 - Why did you pick that option?
 - Why did that option seem like the most [adjective] one?

SECTION: product/ux/ux-research/product-analytics-for-ux-research/_index.md

This page aims to serve as a resource for team members interested in leveraging data we collect about GitLab usage in order to perform or inform user experience research. It provides background information on what GitLab's internal structures are, what data is available and where, and some best practices and tips to help team members get started.

While the information here is meant as a shortcut for people doing UX Research, it only scratches the surface of GitLab's capabilities with data. For more in-depth information, see the Data team's handbook.

Understanding the Product Analytics Landscape

To efficiently get to the usage data you need to conduct research, it's important to understand the various tools and teams involved in data and analytics at GitLab.

Tools

In order of most to least relevant to UX Researchers:

- Tableau - the business intelligence analytics tool we use as a 'frontend' to analyze and visualize data from Snowflake and Service Ping.
- Snowflake - a cloud-based data warehousing platform primarily used for storing and analyzing structured data (i.e., the thing that stores the event data).
- Service Ping - an event tracking tool GitLab built to perform event tracking on self-managed instances (i.e., the thing that tracks events on GitLab self managed).
- Snowplow - an open-source event tracking platform used for collecting, processing, and enriching event-level data from various sources (i.e., the thing that tracks the events on GitLab.com).

Useful Links

- Tableau workbooks (A list of workbooks UX Researchers have found useful)
- Tableau handbook page (The GitLab handbook page on Tableau)
- Analytics relevant to each product stage (A GitLab handbook page for product stage resources)
- GitLab Metrics Dictionary (A website listing all the metrics we collect from Service Ping and Snowplow)
- Data Catalog (Internal handbook page with examples of commonly used data models)
- DBT (Explains the underlying code and column definitions of all GitLab data sets)

The above illustration shows a simplified version of GitLab's usage data stack, based on this document.

Getting Access

Complete an access request to use Tableau. Some UX Researchers have Explorer roles, and some have Viewer roles - if you want to see the differences in capabilities between the two roles, check this handbook page.

To view some dashboards and data, you will need SAFE access, but others are available without. If you do find yourself unable to access information you need, complete a SAFE-related access request for Tableau.

NOTE: Obtaining SAFE access to user data will put you on the designated insiders list, which will require you to obtain pre-clearance in order to sell any GitLab shares you own, as well as restrict the trading window in which you can do so.

Teams to collaborate with

There's a more complete explanation of the various groups that are responsible for data at GitLab, but here are two teams you'll want to know about if you're performing UX Research.

Product Data Insights (PDI) · As the name suggests, this team is responsible for helping folks in Product use data to gain insights. Typically, there is a team member assigned to your area of the product (find your partner on the product data insights team [here](#)).

The Product Data Insights team can take requests (via an issue) for certain kinds of analysis, they can help you find the data you're looking for and are generally a great resource for asking questions during Office Hours or over Slack. They are well-versed in Tableau.

Analytics Instrumentation - This group is focused on building out the instrumentation capabilities of GitLab and its team members, including increased instrumentation coverage across the product and scaling the instrumentation system we use to handle more use cases. This is a group to contact if you run into a block where something you want to do isn't currently available (or you're not sure and the Product Data Insights team doesn't know).

The Analytics Instrumentation team members are the DRIs for data collection across both GitLab.com and Self-Managed. They own Service Ping and Snowplow. They are the ones to go to for questions like:

- How do I instrument a new statistic for self-managed?
- What are the best practices for adding to Service Ping?
- How can I use Snowplow to track a frontend interaction on GitLab.com?
- Can I utilize Snowplow to track events on the server-side?

The state of available information

GitLab has pledged to not use 3rd party software for telemetry, so we have built our own (Service Ping). Due to this and the rapid growth of the product, currently there are gaps in which parts of the product are instrumented for usage analytics and telemetry.

As mentioned above, the analytics instrumentation team has efforts underway to gain more coverage, ensure new features are instrumented before release, and to work on the backlog of non-instrumented features.

Is it possible to associate user level behavior in our GitLab.com data to our Snowplow events (i.e., can we track a single user's behavior)? (Source)

No. Our Snowplow user identifiers are anonymized, while our GitLab.com user identifiers are not. However, it is possible to join Snowplow and GitLab.com data at the namespace (group/project) level (i.e., track more broadly).

Common metrics and segments that GitLab tracks

- Monthly active users (MAU)
 - Examples: MAU per stage, per group, per section
- Performance indicators (PI)
 - Examples: MAU, conversion rate, category maturity advancement
- User segments
 - Examples: Geo Region, of seats, industry vertical
- Product tiers
 - Examples: Free, Premium, Ultimate
- Product sections:

- Examples: Analytics, CD, CI, Dev, Sec
- Delivery
- Examples: SaaS, Paid SaaS, Self-Managed

Working with Product Data Insights team

PDI How to work with us is a great resource on how to open up a product data insight request.

1. Get access to:

- The right tools (usually Tableau online is sufficient)
- The data (do you need SAFE access)

1. Determine the set of questions you're trying to answer, including:

- Which user criteria / segment?
- Which metrics?
- Which components or features (if applicable)?

1. Check with your PM (and/or your PDI counterpart) to determine:

- Is this tracked today?
 - This can be difficult to determine, try asking in data or by working with your PM and PDI counterpart.
 - If not, consider reaching out to the Analytics Instrumentation team or ask in data.
- Is there a dashboard or analysis done already, or do you need a new chart or dashboard?
 - Before opening a request, do a quick search to see if an analysis already exists. If you have trouble determining that, you can ask in data.
 - If new analysis is needed, open a PDI request. Alternatively, you can try to build out your own charts and dashboards, though based on past UXR experiences, partnering with PDI is a great way to get more familiar with Tableau and product usage data.

How can I get help?

- If you ever get stuck or have a question, please ask for help in the data Slack channel.
- Attend the Product Data Insights team's office hours.
- Tableau Training Videos

Example UXR-PDI use case

- Finding quantitative data to triangulate with qualitative research findings (e.g., prioritizing active, commonly used flows; understanding scale and scope of usage).
 - Example: What % of new users are software developers vs. other job titles?
- How many (or what percent) of users (Monthly Active Users, MAU) did X action?
 - Example: How many (and %) of SaaS - Ultimate accounts (not users), used code review features within the last 3 months?
- How many (or what percent) of users went from point A to point B? (i.e., funnel analysis)
 - Example: GitLab .com registration funnel analysis
- How has usage of X action changed over time? (e.g., for adoption of a feature / since a change was introduced)

- Example: Monthly changes in number of Self-Managed, Ultimate accounts that used Code Review features over the past year, broken out by number of seats

Simplified Metrics Glossary

Some terminology used in product analytics dashboards:

- AMAU - Action Monthly Active Users (Unique MAU which engaged with a specific feature)
- CMAU - Estimated Combined Monthly Active Users (Sum of SMAUs across all stages)
- GMAU - Group Monthly Active Users (Unique MAU using features within a specific group)
- SMAU - Stage Monthly Active Users (Unique MAU that used features within a stage)
- Section MAU - Section Monthly Active Users
- UMAU - Unique Monthly Active Users
- DAU/WAU - Daily/Weekly Active Users
- PI - Performance Indicator
- PPI - Primary Performance Indicator (some stages/groups may prefer a different metric than MAU)
- SpU - Stages per user
- SpO - Stages per organization

For more information (or if something's not listed above):

- Data catalog metrics and terms index (internal)
- Product performance indicators (internal)

Useful Tableau Workbooks

There are many dashboards available by searching around in Tableau. Here's a list of a few that UX team members have created or found:

- A list of all dashboards created by the Product Data Insights team
- PDI Dashboards, Analysis, & Insights handbook page
- Firmographic Product Metric Usage Dashboard
- Centralized Product Usage Metrics Dashboard
- Customer Request Dashboard
- Subscription Feature Usage Trends
- UX KPIs
- Snowplow Event Exploration last 30 days
- Service Ping Metrics Exploration Dashboard
- GitLab.com Programming Languages

SECTION: product/ux/ux-research/quantitative-data/index.md

This page defines quantitative data, describes the primary advantages and disadvantages of using quantitative data in UX research, describes best practices, and provides examples for quantitative analysis.

What is quantitative data/research?

Quantitative data is any kind of data that can be quantified through counting or measuring. In contrast, qualitative data is descriptive and not easily distilled into a number. In the context of UX research, quantitative data usually is one of two types:

1. Usage related data, such as clicks on features within GitLab.
1. Survey data of volume (at least a sample size of 30).

What can quantitative data tell us?

Quantitative data can tell us more about the who, what, when, and sometimes how of our product's usage. As an example, quantitative data can answer the following research questions:

- How many users engage with CI Pipeline templates?
- Which GitLab features are used by Executives? (for users who provide their role)

What can't quantitative data tell us?

Quantitative data cannot tell us the "why", because there are always aspects of a user's situation that aren't understood or under our control. For example:

Imagine two users. One is a student who has been following some tutorials they found online to learn how repositories work. The other user is a software engineer who uses GitLab occasionally on their freelance projects. Both users use the same features at around the same time, and neither have their profile information filled out.

From this point of view, there is no way to tell the difference in use cases through usage analytics, and over time these users will end up being misrepresented in the product. While quantitative data is powerful, it's important to recognize the limitations of what it can provide.

When to consider using quantitative data?

If your hypotheses and research questions are similar to the examples below, then quantitative data may be one tool you consider using. Remember, this doesn't mean that quantitative data is the only tool to use, because there are still limitations as to what analytics can tell us.

Examples of questions that point towards using quantitative data:

- What do users click on when they are on a certain page?
- How many users begin and finish a process?
- How does the usage of one feature compare to another?
- When is a certain feature used?
- How often is a feature used?
- Which different types of options do users select?

You can usually spot these questions, as they tend to start with "what," "how," or "when." Here is a walkthrough of how designers are using data to make decisions.

What is the process of conducting quantitative research?

The typical process is as follows:

1. Form questions and hypotheses that can be answered by quantitative research
1. Collect data
1. Interpret/analyze the data
1. Summarize findings

Why do I need to generate questions and hypotheses prior to conducting quantitative research?

Before starting any kind of quantitative research, it is important to identify your hypotheses and questions to avoid the problem of "data snooping." In data snooping, a lack of defined hypotheses/questions leads you to find results that seem meaningful, but are actually misleading and misrepresentative of the user population. Hypotheses/questions serve as an anchor for your overall analysis.

How do I collect data?

At GitLab, there are two main approaches to collecting data: 1) querying from usage data in Snowflake or Tableau - see this handbook page for more information on GitLab's data sources - and/or 2) using survey data.

You can use these two data sources to build a comprehensive understanding of your problem. For example, you might use usage data to first understand the user population you are looking at and then identify where you will need to go to get more data. Once you understand your user population, you might use a survey to uncover additional insights.

How do I look at/interpret quantitative data?

One way to look at data is with data visualizations. Data visualizations are graphical representations of data, commonly seen as charts or tables. Data visualizations are used to represent a trend in your data, or display a summary of your data.

There are many types of visualizations, and each one may suit a particular situation better than another. Below is an overview of common visualizations using example charts driven by SUS data, but you may also want to visit this helpful blog post for additional examples.

- Tables are good to use when you want to summarize a portion of your data or display aggregations.

- Pie charts are used to compare data in different categories.

- Bar charts are used to compare data across distinct categories, like fiscal quarters.

- Line charts can be used to see trends. Time series are a type of line chart where the x-axis is always time.

- Scatter plots are used when comparing two numeric variables. These charts can have some of the most flexibility, but that also comes with the risk of creating overly complex visualizations.

What should you avoid when creating visualizations?

There are dozens of ways to lie with visualizations, even by accident. Some tips when creating your visualizations:

- Visualizations should be quick and easy to interpret. The visualization should not result in more questions than answers.
- Each of your axes should be understandable and practical:
 - Always start your axis at 0.
 - Do not skip or remove large sections of your data by limiting the axis.
 - Do not use double Y axes unless absolutely necessary.
 - If you are comparing two graphs, ensure the axes are on the same scale and range.
- For chronological studies, always put time on the X axis.
- Keep it simple. Titles should be clear and concise, and there should be enough context for a team member unfamiliar with the topic to understand the message.
- State the limitations, including the date the data was pulled, and where it was pulled from. Use subtitles for references when needed.
- Consider your audience's familiarity with the data set. A simple and informative graph can often have more impact on your audience than a "flashy" graph with multiple groupings and axes.

Summarizing quantitative data

This section covers how to read visualizations, look for trends, and turn the results into insights. Start by looking for any kind of patterns. Some examples are:

- Cyclic patterns based on time
- Similarities or differences when comparing across one category or more
- Data that is consistently increasing or decreasing.
- Two variables that are correlated with each other. The correlation could look like:
 - As one metric goes up/down, another metric consistently goes up/down with it.
 - As one metric goes up/down, another metric consistently does the opposite.

Once you have identified patterns, summarize those patterns as best you can while considering your original hypotheses/questions.

An example of quantitative data summary

The following is an example of using quantitative data to investigate Actionable Insights (AI). The research question was, "What are the potential factors in whether an (AI) issue is successfully completed or not?" First, the researcher pulled usage data in Sisense related to GitLab issues with the Actionable Insight label. Then, they created these charts:

Based on these charts, it is evident that there is a huge decline in issues closed after 7 months. So the findings were presented as, "If Actionable Insight issues are not addressed within 7 months of creation, they are likely to be left open."

Additional examples of Product Designers using data for decisions

State the limitations of your data

When sharing your research, state what you believe are the limitations that may have impacted your results. For example, if your data:

- Is missing a collection of users
- Includes usage from only the last 3 months
- Has concerns about its accuracy
- Does not directly measure your research topic

Be sure to include a description of the limitations in your summary. This information is just as relevant to the research as the metrics, and it provides context to the data. In some cases, you may decide it is not worth presenting the data, because of multiple problems.

Best Practices

Using quantitative data to track the success of a feature

Tracking key metrics that are formed through analytics can help you understand how a feature is performing in production over time.

Consider adding analytics to features as they are developed and improved. This will enable you to track how successful the design or new feature was.

Using quantitative data alone

There are some instances where you can use quantitative data alone. Some examples are:

- Are users finding a new feature?
- Which pages are visited the most?
- Which browser types do users use to access our application?
- Are more users accessing a page now compared to last quarter?

While the examples above can be answered solely with quantitative data, pairing with qualitative data will provide you with a more complete analysis. That's because qualitative data helps you to understand the "why" behind what's happening.

The perfect combination: quantitative and qualitative data together

Some of the most useful research designs come from mixed methods. Both quantitative and qualitative methods have their limitations, but you can use them together to answer complex questions and provide a more complete story. By combining methods, you are able to answer both the "why" and the "how" or "what" of the problem.

Example 1: Understand how users interact with a specific page.

1. Use analytics to find out how many users visited a specific page and which features they used on that page. Focus on general characteristics of those users (account age, stages they visit, and so on).

1. Launch qualitative research that includes personas with any chance of overlapping with your

page/feature. Be sure to expand your sample to more than just your analytics, remembering that our analytics aren't always comprehensive (for example, we have limited data on Self Managed users). Use subject matter knowledge to see the possible gaps in your data, making sure to advocate for the hidden use cases.

1. Use interviews or walkthroughs to understand user motivations.
1. Incorporate your enhanced understanding in the proposed solution.

Example 2: Finding areas for improvement of a new feature for future iterations.

1. Monitor usage analytics with the feature flag after launching the new feature. Examine things like time on page and how many users interacted with other features during the workflow.
1. Understand high-level goals by interviewing users with personas that have any chance of interacting with the new feature, as well as any persona identified in the usage data.
1. Pair usage analytics and the interview insights to visualize workflows that interact with other features.
1. Include priorities based on user input and time-on-page trends.
1. Use those insights to help map future iterations in the roadmap.

What do you do when you don't have enough quantitative data?

There are many common scenarios where you might not have enough data. For example:

- A new metric was implemented, but you only have a month's worth of data.
- Data is only reported from SaaS users and not self-managed.
- There was an error/bug in data scrubbing, and you are missing a period of time.

There are a couple of different ways you might address this.

(1) Expand research by supplementing your quantitative data with qualitative research. For example:

1. During a quantitative investigation of a navigation element, you see an incredibly small number of users who consistently click on a different element than other users.
1. After more investigation, you connect several character demographics and possible user personas to this user group. However, this research is unable to answer the question of why those users behave the way they do.
1. To attempt to answer this, you launch user interviews using the personas you found. After summarizing the interviews, you see that the users had a small set of overlapping use cases due to their shared industry, which leads to a newly identified JTBD for those personas.

(2) Continue forward with the current product roadmap while summarizing your insights and the possible shortcomings. For example:

- During problem validation, you find some data that points in the opposite direction of your hypothesis and assumptions and may contradict parts of your product roadmap. However, the data has a low sample size and perhaps leaves out some context in the workflow.
- When writing the results, speak to the facts of the data and your possible assumptions or biases. Explain how the data does point to one direction, but there is not enough evidence to suggest changing the roadmap.
- Additionally, determine whether the insights are worth preserving and investigating more if

an opportunity arises.

It is NOT recommended to fill your analysis with data from another project. This can create problems due to sample sizes from different populations making the data look different than reality.

SECTION: product/ux/ux-research/recruiting-participants/_index.md

At GitLab, our UX Research Operations Coordinators work closely with those requesting research to make sure the people they are gathering feedback from are the right ones. If you are filling in for a UX Research Operations Coordinator, use this page as a reference.

How to recruit participants when you aren't sure where to find them

1. Identify your target audience (i.e. what characteristics should they have?)
 1. Craft your screener
 1. Create a Calendly event
 1. Open a recruitment request issue
 1. Open an incentives request issue

How to recruit participants who are customers you know

1. Identify your target audience (i.e. what characteristics should they have?)
 1. Reach out to your customers and ask for their interest
 1. Coordinate directly with your customers to schedule their session. This may include creating a Calendly event
 1. Open an incentives request issue

How to recruit participants from issues

1. Identify your target audience to determine what characteristics they should have
 1. Craft your screener
 1. Create a Calendly event
 1. Solicit participants by posting a link to the screener on the related public issue
 1. Open a recruitment request issue
 1. Open an incentives request issue

A UX Researcher or UX Research Operations Coordinator may collaborate with you throughout the process, and we're always available to answer questions.

Identify your target audience

Collecting the data that will help you answer your questions is essential. The only way to do that is to ensure you speak with the right people. Ask yourself the following questions to help define your target audience:

- What research methodology have you chosen?
- How many participants do you need?

- Who do you want to hear from/speak with?
- What are the tasks your participants should be doing?
- What are the tools your participants should be using?
- What is their level of experience?

The answers to these questions should flow directly from the goals and objectives that you've previously identified for your study. It's important that everyone involved in the research study agrees on these and that they're finalized before we start recruiting. It's better to start with clear objectives than to try to shoehorn participants into a study for which they simply aren't a match.

It's best to think of tasks and responsibilities instead of job titles, because often a person's job title doesn't fully correspond to the tasks they perform.

Craft your screener

The screener has a specific function - it's meant to identify the people who are your target demographic, so that you can ask them the things you really want to know in the study. A best practice is to copy the Screener Template (internal link only) in Google Docs and collaborate with your stakeholders until all questions are finalized. For tips on how to write an effective screener, check out this page. Then, you'll create a screener in Qualtrics that includes the questions. If you don't have access to Qualtrics, request it. Recruiting will not begin until the finalized screener is created in Qualtrics and shared with the UX Research Operations Coordinators.

Your questions on the screener must match your participant criteria. This allows the UX Research Operations Coordinator to review your desired criteria and know which answers you want to see on the screener. The more abstract or open-ended you get in the screener, the harder it is for UX Research Operations Coordinators to parse which answer it is you're looking for. A best practice is to avoid using open-ended questions in screeners.

Participants must agree to each of these questions to take part in moderated studies. You must include these questions in every screener:

- Consent to record the session
- Consent to store the recording in GitLab

Determine if you need the IP Assignment and/or GitLab's Individual Contributor License Agreement

- Learn more about the IP Assignment and when to show it

Common questions we include are:

- Job title
- Common tasks
- Industry
- Team and company size
- Tools used

There are copies of these questions with standard phrasing available in the UX Research library in Qualtrics. There is also a generic template in Qualtrics based on the screener template noted above. To use this template, create a new project in Qualtrics and select the following options:

- How do you want to start your survey? Use a survey from your library
- Library UX Research & Product
- Survey UXR Generic Screener (from the Templates folder)

After selecting Create project you will be redirected to the survey builder within Qualtrics and will be able to edit the screener however you see fit.

Sometimes, we need to recruit participants from the same organization. In these cases, consider adding a question at the end of the screener to ask potential participants to recruit additional prospects (called snowball sampling). Example:

> We are also interested in talking to people from the same organizations/teams, but who have different roles than yourself. Everyone's interviews will be done separately and everyone will be paid the same amount, once their session is over. Your participation in the session is not contingent on anyone else participating. If you know of any colleagues who would be interested in taking part in this study, please provide their email and their role/job title, and we will send them a link to this screener.

Sometimes, we are able to combine efforts by using a common screener across different studies. This is called a Common Screener.

It's important to remember that a screener is not the same as a survey. There can be a temptation to pile on extra questions that you want to know the answer to, but aren't strictly necessary for determining whether that participant should get through the door. You will benefit from including these questions in your discussion guide or script instead.

Create a Calendly event

To schedule participants, you will need to create a Calendly event. The UX Research Operations Coordinator will send individual emails to qualified participants and ask them to book a time with you. Set up Calendly [here](#).

1. Click 'New Event Type'.
1. Choose 'Create' for the One-on-One meeting option.
1. For the Event name, include the topic of the study.
1. For Location, enter a Zoom meeting URL. A best practice is to use a recurring meeting.
1. The Description/Instructions are optional.
1. For Event link, use a short phrase to describe the topic of the sessions.
1. For Event color, choose a color you're not currently using.
1. For Event Duration, reference your research method. Typically our user interviews and usability sessions run from 30-60 mins.
1. For Date Range, click 'Edit' and choose 'Over a Date Range' from the drop down menu. Then choose the days you wish to hold your sessions. A best practice is to give yourself 5-7 business days to conduct research sessions.
1. You can click on each day in the calendar to indicate the time(s) you are available for

sessions.

Open a recruitment request issue

Prior to opening a recruitment issue, please have all materials ready:

- Screener
- Script
- Recruitment timelines

Without these materials the UX research operations coordinator will not be able to move forward with your request. When prematurely creating recruitment issues it will over-inflate our tracking on turn around time for recruitment issues.

Step 1 - Open a Recruitment Request Issue

This step provides us with all the necessary information we need to complete your request.

- Navigate to the UX Research Project and open a new issue.
- Select the <Recruiting Request> template and fill in the information required.

Step 2 - Wait one-two working days

The UX Research Operations Coordinator will claim the study within one-two working days and work with you to find your participants. They will reach out to you with any further questions and an update on their timeline. When your study is claimed, the coordinator will remove the ReOps::Triage label. It will be replaced to identify they are working on it e.g. ReOps::Cait - for Caitlin Faughnan.

What if I want to get a head start, but I'm not quite ready?

If your study is not yet ready for recruitment and you're opening the request early, please start your title with WIP or DRAFT.

The UX Research Operations Coordinator will claim the study but not work on the request until the issue leaves the WIP/DRAFT phase.

When the issue is ready to leave the WIP/DRAFT phase, @mention the UX Research Operations Coordinator assigned to your request.

What will the UX Research Coordinator be responsible for on my request?

The UX Research Operations Coordinator is responsible for launching recruitment campaigns, prioritizing requests, scheduling participants - if required, reimbursing them for their time, and ensuring NDAs, as needed, are signed and stored properly.

What recruitment method will be used?

The UX Research Operations Coordinator assigned to your project will select the best method for recruitment, based on the recruitment criteria. That may include using multiple approaches, depending on the criteria. Options include:

1. Data Warehouse
1. Marketo
1. Respondent.io - a third party recruitment tool
1. Social Outreach
1. Direct Sourcing via LinkedIn
1. Docs Site Banner
1. UserTesting.com - a third party recruitment tool (primarily used by Product Designers to self-serve)

Learn more about these recruitment methods.

Scheduling participants

Once the screener has been sent to potential participants, the UX Research Operations Coordinator will monitor the results and share them with the person who requested the recruiting. This is done via a Google spreadsheet. They will suggest which potential participants best fit your recruiting criteria. The DRI for the research may also provide a list of names to the UX Research Operations Coordinator from the responses that they would like invited.

Once potential participants are agreed upon, the UX Research Operations Coordinator will send the scheduling email to qualified participants via Gmail. The UX Research Operations Coordinator relies on you to update them in a timely manner as to whether any bookings have been made. You should plan to alert them within a couple of business days if no bookings have been made on your calendar, so they can send reminders or fresh invitations. They will repeat this cycle until the request is fulfilled.

Reimbursing participants for their time

The UX Research Operations Coordinator assigned to your request will handle paying the participants. However, it is up to the DRI on the project to let the coordinator know within 48 hours of session completion to ensure we are providing a positive experience for our participants by processing their incentives within a timely manner. When all participants have been paid and there's no outstanding sessions remaining, the UX Research Operations Coordinator will close out your issue.

When do I submit an Incentives Request with UX Research Operations?

Since you created a Recruitment issue to kick off your process, this is automatically done; as this step is already built in as part of the process. That means you don't have to do anything and the incentives will automatically be handled by the assigned UX Research Operations Coordinator.

What if I found my own participants and just need to hand out incentives?

If you have a scenario where you didn't create a Recruitment issue and need to distribute incentives, you'll need to open an Incentives Request issue.

- Navigate to the UX Research Project and open a new issue.
- Select the Incentives Request template and fill in the information required.

- Paste the link to the spreadsheet in the issue.

One of the UX Research Operations Coordinators will pick up your request within one business day and process the incentives for your participants, provided all the required information is in the spreadsheet.

If your study is ongoing, simply comment on the issue or tag the UX Research Operations Coordinator in the spreadsheet when you have more incentives that need to be processed.

Recruitment timelines

Recruitment typically takes 2 weeks for target participants who are plentiful in our research panel and customer database, such as Engineers and DevOps professionals.

Recruitment for low-incidence users, like security professionals or users of a newly released feature, may take more than 2 weeks.

In rare cases, when your target participants cannot be found, the UX Research Operations Coordinator will suggest other options.

In all cases, the UX Research Operations Coordinator will keep you updated on the progress of the recruitment effort by commenting on the recruiting issue.

Promotional games

If you are planning to recruit users through a promotional game or contest (e.g., Opportunity to win 1 of 3 \$30 (or equivalent currency) Amazon Gift cards), please review the following information in the handbook and consult with legal. For information on contacting legal, please refer to how to reach us in the Legal Team handbook page. Engag